Midterm 2 Practice Problems

MATH 4400/6400 Fall 2026

Give complete justifications, including the relevant hypotheses when applying a theorem. You may quote results proved in class unless a proof is requested.

I. The division algorithm in quadratic rings

1. The division algorithm in $\mathbb{Z}[\sqrt{-2}]$

Find $\gamma, \delta \in \mathbb{Z}[\sqrt{-2}]$ such that

$$5 + 4\sqrt{-2} = (3 + \sqrt{-2})\gamma + \delta, \qquad N(\delta) < N(3 + \sqrt{-2}).$$

2. The Euclidean algorithm in $\mathbb{Z}[i]$

(a) Use the Euclidean algorithm in $\mathbb{Z}[i]$ to compute $\gcd(7 + 5i,\, 3 + 7i)$ up to multiplication by a unit.

(b) Express your gcd as an $\mathbb{Z}[i]$-linear combination of the two given elements.

II. Norms, divisibility, and prime elements

3. A rational prime in $\mathbb{Z}[\sqrt{-2}]$

Use the norm to prove that 5 is a prime element of $\mathbb{Z}[\sqrt{-2}]$.

4. Norm for $\mathbb{Z}[\sqrt{d}]$

Let d be a nonsquare integer and let $\alpha = x + y\sqrt{d} \in \mathbb{Z}[\sqrt{d}]$ be a nonzero element with $\gcd(x, y) = g$. Prove that $g^2 \mid N(\alpha)$ and that for every $m \in \mathbb{Z}$, $\alpha \mid m \iff (N(\alpha)/g) \mid m$.

III. Units and prime factorization

5. Units in $\mathbb{Z}[\sqrt{2}]$

Show that $7 + 5\sqrt{2}$ is a unit of $\mathbb{Z}[\sqrt{2}]$ and compute its inverse.

6. Prime factorization in $\mathbb{Z}[\sqrt{2}]$

Find a prime factorization of $68 + 68\sqrt{2}$ in $\mathbb{Z}[\sqrt{2}]$ and justify your answer.

7. A failure of unique factorization in $\mathbb{Z}[\sqrt{-6}]$

Find two essentially different factorizations of 10 into primes in $\mathbb{Z}[\sqrt{-6}]$.

8. A failure of unique factorization in $\mathbb{Z}[\sqrt{15}]$

(a) Prove that $2, 3 \in \mathbb{Z}[\sqrt{15}]$ are primes.

(b) Prove that 6 admits two essentially different factorizations into primes in $\mathbb{Z}[\sqrt{15}]$.

IV. Continued fractions and Pell's equations

9. Continued fractions and a Pell equation

(a) Compute the simple continued fraction expansion of $\sqrt{6}$.

(b) Use part (a) to find the fundamental solution (x, y) of $x^2 - 6y^2 = 1$.

(c) Starting from the fundamental solution, construct two further positive integer solutions.

(d) Does $x^2 - 6y^2 = -1$ has any integer solutions (x, y)?

10. A negative Pell equation and its square

(a) Find the simple continued fraction expansion of $\sqrt{10}$.

(b) Use part (a) to find the fundamental solution (x, y) of $x^2 - 10y^2 = -1$.

(c) Use part (b) to find the fundamental solution (x, y) of $x^2 - 10y^2 = 1$.

(d) Find all units in $\mathbb{Z}[\sqrt{10}]$.

11. A family of Pell's equations

Let $r \geq 2$ be an integer and put $d = r^2 - 1 \geq 3$ or $d = r^2 + 1 \geq 3$.

(a) Prove that d is not a perfect square.

(b) Prove that $r + \sqrt{d}$ is the smallest element > 1 of $\mathbb{Z}[\sqrt{d}]$ having norm 1. Then deduce all positive integer solutions (x, y) of $x^2 - dy^2 = 1$.

V. Diophantine equations

12. A variant of the Pythagorean equation

Find all integer solutions (x, y) to $x^2 + 2y^2 = z^2$.

13. Squares and cubes

Find all integer solutions (x, y) to $x^2 + 1 = y^3$.

VI. The infinitude of primes

14. Primes modulo 6

(a) We proved in class that there are infinitely many primes $p \equiv 3 \pmod 4$. Adapt the proof to show that there are infinitely many primes $p \equiv 5 \pmod 6$.

(b) Suppose that $p \neq 3$ is an odd prime. When is -3 R p and when is -3 N p?

(c) Show that if $p \equiv 5 \pmod 6$ and $x^2 + 3y^2 \equiv 0 \pmod p$ is solvable, then $p \mid x$ and $p \mid y$.

(d) Adapt our proof of the infinitude of primes $p \equiv 5 \pmod 8$ to show that there are infinitely many primes $p \equiv 7 \pmod{12}$. In particular, there are infinitely many primes $p \equiv 1 \pmod 6$.